

Muhammad Farhan Azmine

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Objective

Seeking an internship position in Digital Design/DV or CPU/GPU Engineering

Education

Virginia Tech (GPA: 4.00 / 4.00)

PhD in Computer Engineering (Direct PhD; MS completed)

Blacksburg & Alexandria, VA

Aug 2022 – Dec 2026 (Expected)

- **Relevant Coursework:** Advanced Digital Design, Advanced Computer Architecture, Testing VLSI Techniques, VLSI Device Modeling, Advanced Analog IC Design, Deep Learning, Advanced Machine Learning, Computation in Data Science

Publications

- **Muhammad F. A.,** Li, R., Sharma, G., & Yi, Y. (2025). SpikeSpec: On-Chip Learning Neuromorphic Accelerator for Spectrum Sensing. *IEEE Transaction CAD*, Feb 2025.
- Li, R., **Muhammad F. A.,** Sharma, G., & Yi, Y. (2025). Efficient Digital Architecture of Spiking Encoders. In *Proc. ISQED 2025*.
- Lin, C., **Muhammad F. A.,** Liang, Y., & Yi, Y. (2024). Neuro-Inspired AI Accelerator for 6G Networks. *Front. Comput. Neurosci.*
- Lin, C., **Muhammad F. A.,** & Yi, Y. (2023). Accelerating Wireless Communications with FPGA-Based AI. In *ICCAD 2023*.

Research & Design Experience

Research Assistant (Functioning as RTL Engineer – Complex Digital System Design & Verification)

June 2023 – present

Supervisor : Dr. Yang (Cindy) Yi

RISC-V 5 stage Pipelined CPU Tapeout on Sky130 PDK with Genus & Formal Verification Framework with JasperGold

[\[Github link\]](#)

Fall 2025

- Designed and verified using SystemVerilog a **5 stage pipelined RV32I RISC-V CPU** with forwarding circuit to solve hazard, taped out on **Sky130 PDK** (21.8K cells, 334.9K μm^2 total area, 15.9 mW total power), meeting timing across FF/SS/TT corners in **Cadence Genus**.
- Developed a **UVM-lite environment with abstracted register/memory monitors** and **property-based formal checks** for **40+ RV32I instructions** using **Cadence JasperGold** and **Xcelium**, achieving **100% proof convergence**, **98% coverage**, and **80% debug effort reduction**.
- **Reduced task flow by 80%** by creating a **Makefile script** to automate **C code** disassembly into **LLVM IR** for **R32IV CPU** using the **riscv64-gcc-elf** toolchain in **Ubuntu Linux 24.04 OS**.

RTL Design and Verification of Direct-Mapped Cache Memory with RAM and Control Unit using Xilinx Vivado

[\[Github link\]](#)

Fall 24

- Designed and verified in SystemVerilog an integrated direct-mapped cache memory system comprising cache memory, RAM, and a control unit, with a 15-bit address input for a 128-bit cache line and 1024 entries.
- Ensured **100% efficient cache hit/miss synchronization** using read/write handshake signal verification and an FSM-based controller.
- Performed cache hit/miss determination through comparison of tag bits and valid bit.

PPA optimization of Spiking Neural Network RTL architecture design with On-Chip learning for Spectrum-Sensing

[\[Github link\]](#)

Fall 22 - Spring 24

- Achieved **100% SystemVerilog Post-Si validation** through hardware–software co-integration on **Zynq CPU**, enabling real-time sample delivery to **RTL (PL)** and result capture via **ZynQ AXI-UART (PS)**, with a **custom C++ parser** for CSV input processing and output generation.
- Reduced **RTL area by 60%** in **SystemVerilog** using resource-shared adders and LUTs via **serialization with SIPO shift registers**.
- **Throughput increase by 58 MHz** with critical path balancing between **priority encoder & exponential approximator**
- **Reduced latency by 50%** using simple **dual port memory ram** in read-then-write mode for weight learning update

RTL integration of Ethernet-MAC IP interface with AI Accelerator inference chip for OFDM symbol detection

[\[Github link\]](#)

Spring 23-Fall 23

- **Boosted design frequency by 100 MHz** by implementing **Clock Domain Crossing** to achieve 200 MHz frequency for target accelerator through synchronizing with Ethernet PHY communication at 125 MHz using **Ping-pong buffer**, **CDC AXI-handshake IPs** and **Asynchronous FIFOs**
- **Improved SystemVerilog verification coverage by 30%** through modifying BIST testbench in frame data transfer between Ethernet-PHY and target accelerator by creating **over 4 protocol-variant Ethernet frame stimulus patterns** including error-injection & backpressure scenarios
- **Reduced IP area usage in SystemVerilog RTL by 33.3%** through **DSP48E1 IP** integration at RTL-level for inference MAC operation

Technical Skills

- **Techniques:** RTL design, STA, PPA, Clock-Domain-Crossing, UVM, DFT, FPGA-IP Integration, ASIC implementation
- **Languages:** SystemVerilog, Verilog, Matlab (Script), Python (OOP) [\[Github link\]](#), C++, Java [\[Github link\]](#), Tcl, Linux Shell
- **Tools:** Cadence Suite (Xcelium, Genus & JasperGold), TensorFlow [\[Github link\]](#), PyTorch, Scikit-learn [\[Github link\]](#), Vivado, Quartus
- **Concepts:** UVM, SVA Formal Verification, AXI-DMA, AXI4, UART, Ethernet-TCP/UDP, SPI, VGA, RISC V ISA, Cache memory mapping